



ADI *Power*

Power Management Fundamentals II Session 2: Explore Boost and Buck- Boost Converter Operation

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Agenda

- 1 Overview
- 2 Boost Regulators
- 3 Buck-Boost Regulators

Powering Up – Module Overview

Session 1 LDOs & Bucks	Session 2 Boost & Buck-Boost	Session 3 Sepic, Ćuk & Inverters	Session 4 Isolated Converters	Session 5 Power Path, Hot-swap & Surge-Stopper	Session 6 Layout Considerations
LDO Regulators Buck Regulators	Boost Regulators Buck-Boost Regulators	Sepic/Ćuk Inverting Regulators	Flyback Converters Forward Converters	Power Path Hot-Swap Surge-Stopper PoE	Layout
Takeaway A Deeper Look into the workings of LDOs & Bucks	Takeaway A Deeper Look into the workings of Boost & Buck-Boost	Takeaway A Deeper Look into the workings of Sepic/Ćuk & Inverters	Takeaway A Deeper Look at Isolated Converters	Takeaway A Deeper Look into Power Input Circuits	Takeaway A Deeper Look into Layout Considerations

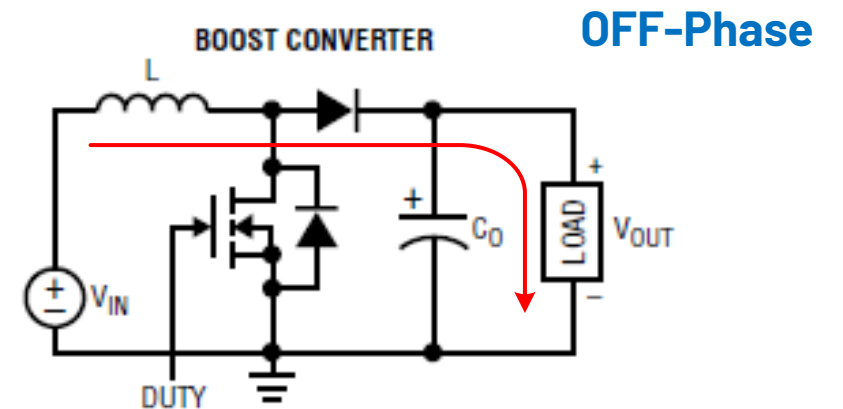
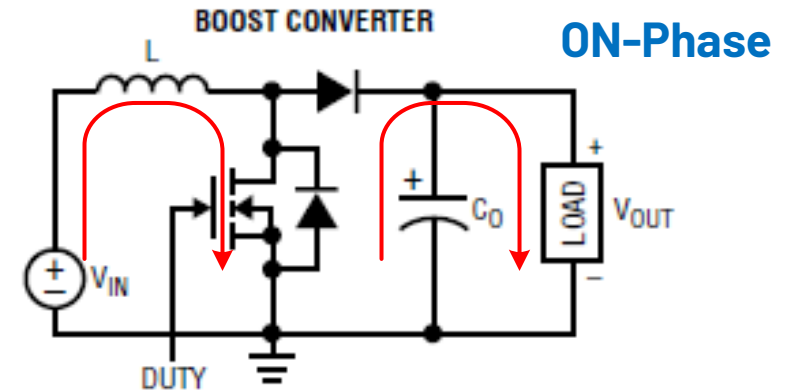
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Boost Converters

Boost converters

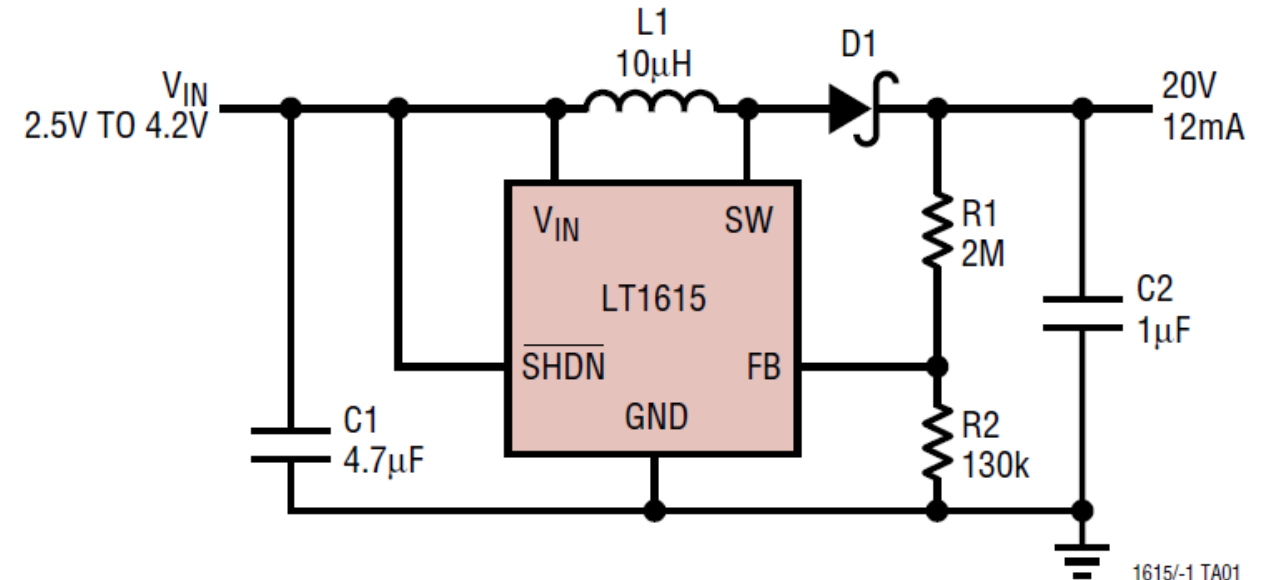
- $V_{OUT} > V_{IN}$
- ON-Phase: Primary switch conducts current to GND. During this time, C_{OUT} provides all load current
- OFF-Phase: Primary switch is OFF, inductor provides output current and re-charges C_{OUT}



Boost Converters

Boost converters

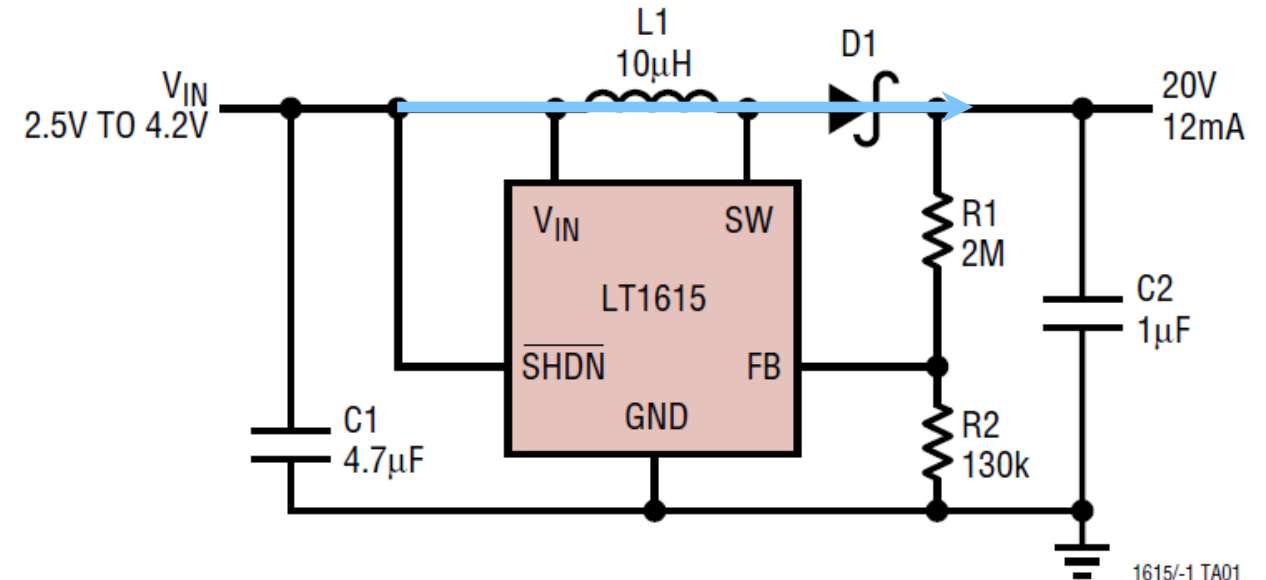
- $V_{OUT} > V_{IN}$
- By conservation of power, Inductor current is greater than I_{OUT}
- $I_L \approx \frac{V_{out} * I_{out}}{Eff * V_{in}}$ (average)
- Switch current rating must be greater than I_L plus $\frac{1}{2}$ Inductor ripple current
- Duty Cycle = $\frac{V_{out} - V_{in}}{V_{out}} = 1 - \frac{V_{in}}{V_{out}}$
- LTSpice Example



Boost Converters

Boost converters

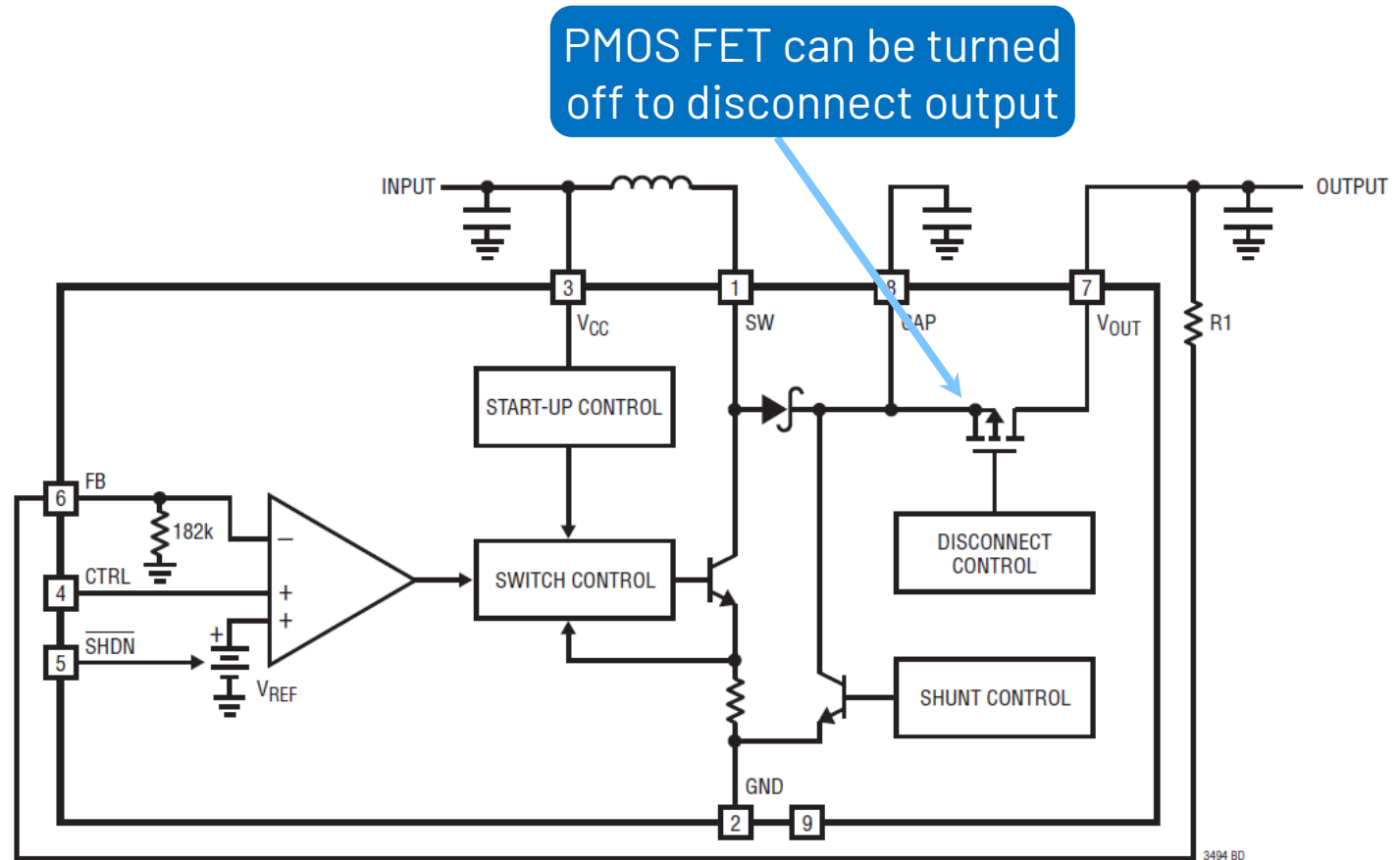
- Output current path
- Note that even when the converter is OFF, there is a path from V_{IN} to V_{OUT} through the inductor and diode
This can be an issue if there is a fault on the board which shorts V_{OUT} to GND



Boost Converters

Boost converters

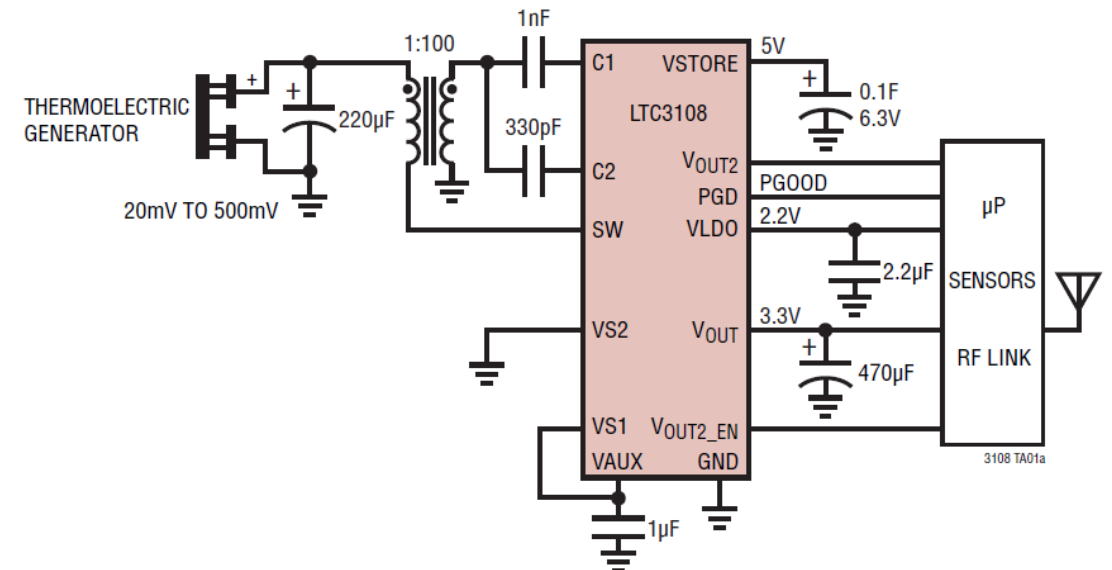
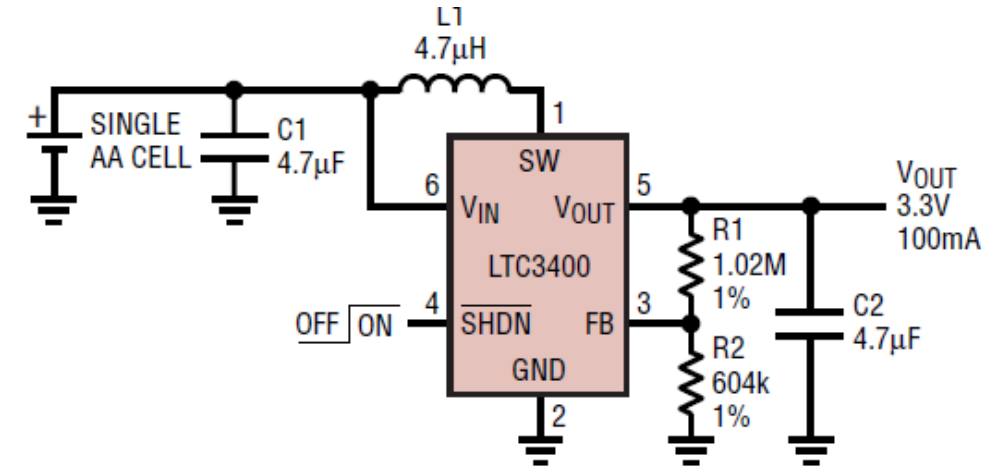
- Output disconnect
- Boost converters like the LT3494 include an output disconnect function to block the path from V_{IN} to V_{OUT}



Boost Converters

Boost converters

- Minimum V_{in}
- Some Boost converters can start up from very low input voltages. Many are available with start-up of 1.0V or less (see LTC3400).
- Many can run down to lower input voltages (e.g. 400mV) once up and running
- A few can start up even lower V_{in} (see LTC3108)
- Often, low-voltage Boost converters will get started at a low voltage and then run off V_{OUT} once the output is greater than V_{IN}



Boost Converters

Boost converters

- Boost Controllers
 - For higher power levels
More precise Loop control
 - Example: $V_{IN} = 12V$, $V_{OUT} = 24V$ @5A
 - LTPowerCAD Example

Boost Converters

Boost converters (monolithic)

Definitions

- Continuous Conduction Occurs at heavy loads, inductor current is always > 0
- Pulse-skipping Mode
When inductor current falls to zero, pulses are skipped to keep the output in regulation while not allowing negative inductor current.
Most of internal circuitry is kept in the ON state, so the part can react quickly when switching must resume.
- Better light load efficiency, but with low output voltage ripple

Boost Converters

Boost converters (monolithic)

Definitions

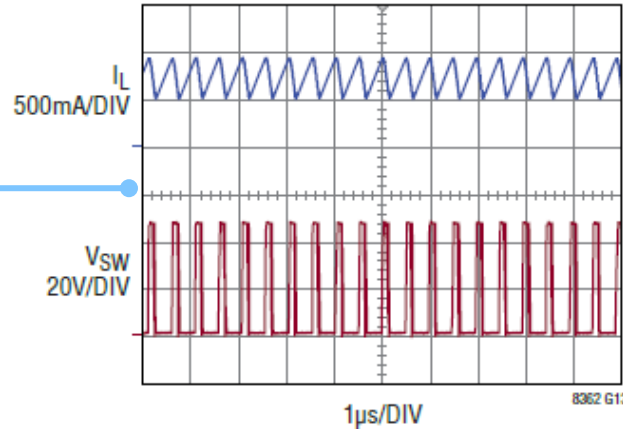
- Burst Mode
When load current is quite small, the part may shut down all internal circuitry except the reference and output voltage comparator. The part sleeps until V_{OUT} drops below a preset threshold, when switching resumes.
- This mode offers the best light-load efficiency, but the V_{OUT} ripple and transient response may show larger deviations than other modes

Boost Converters

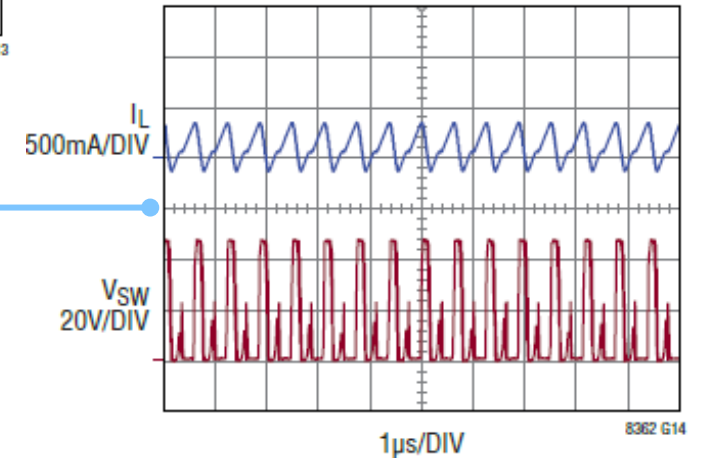
Boost converters (monolithic)

- Efficiency considerations
- Continuous Conduction
Occurs at heavy loads, inductor current is always > 0
- Pulse-skipping Mode
Better light load efficiency, but with low output voltage ripple
- Burst Mode
Best light load efficiency, with somewhat higher V_{OUT} ripple

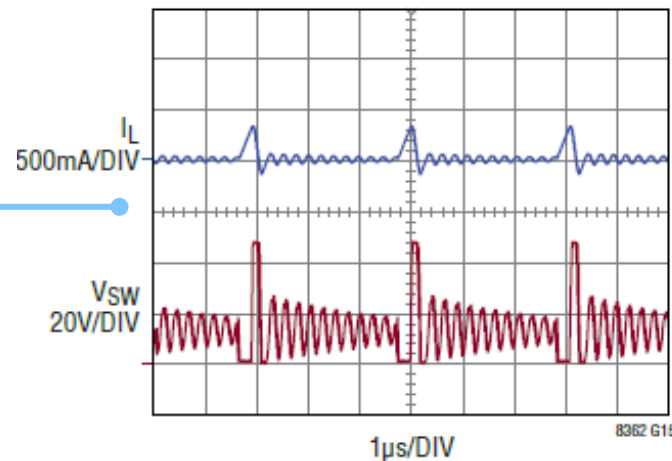
Switching Waveforms
(in CCM)



Switching Waveforms
(in DCM/Light Burst Mode)



Switching Waveforms
(in Deep Burst Mode)



Boost Converters

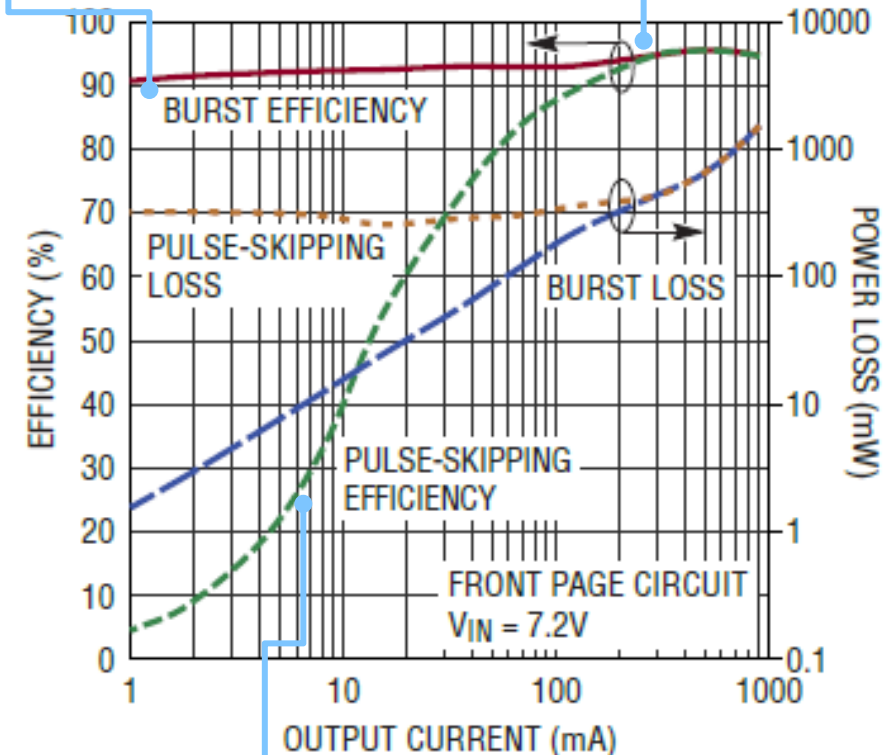
Boost converters (monolithic)

- Loop Control
There are Voltage Mode and Current Mode Boost converters
- Efficiency is enhanced via Pulse-Skipping and/or Burst Modes at light load
- Some monolithic Boost converters automatically transition from PWM Mode to Burst Mode. These are mostly Current Mode control parts.
- Example: LT8337

Burst efficiency

Efficiency and Power Loss vs Output Current

Transition to PWM at high load current



Pulse-Skipping
Efficiency

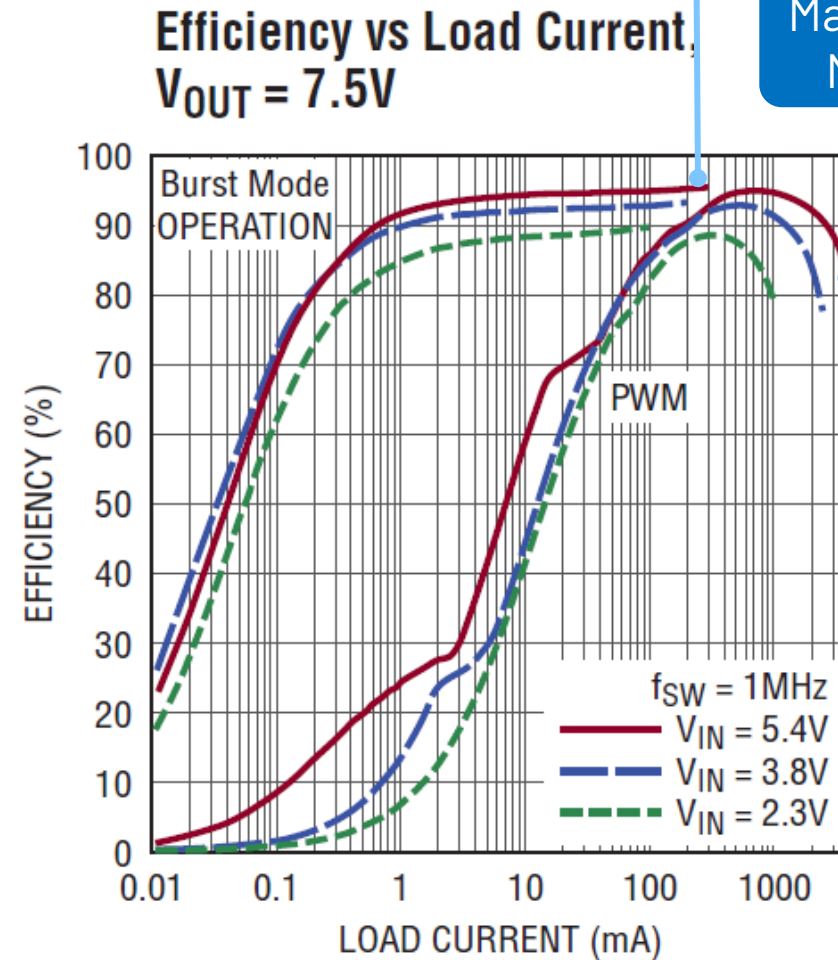
Boost Converters

Boost converters (monolithic)

- Loop Control
- Other monolithic converters require manual intervention (PWM pin) to select Burst or PWM. These are primarily Voltage Mode parts and the full output current isn't available in Burst Mode
- Read the Datasheet!
- Example: LTC3124

Reduced I_{OUT} with
Burst Mode selected

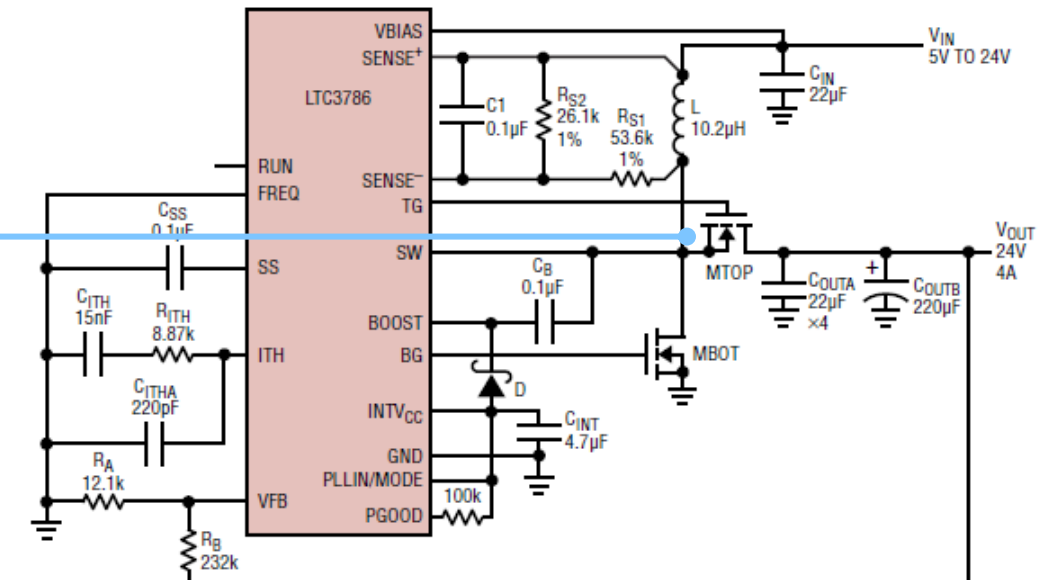
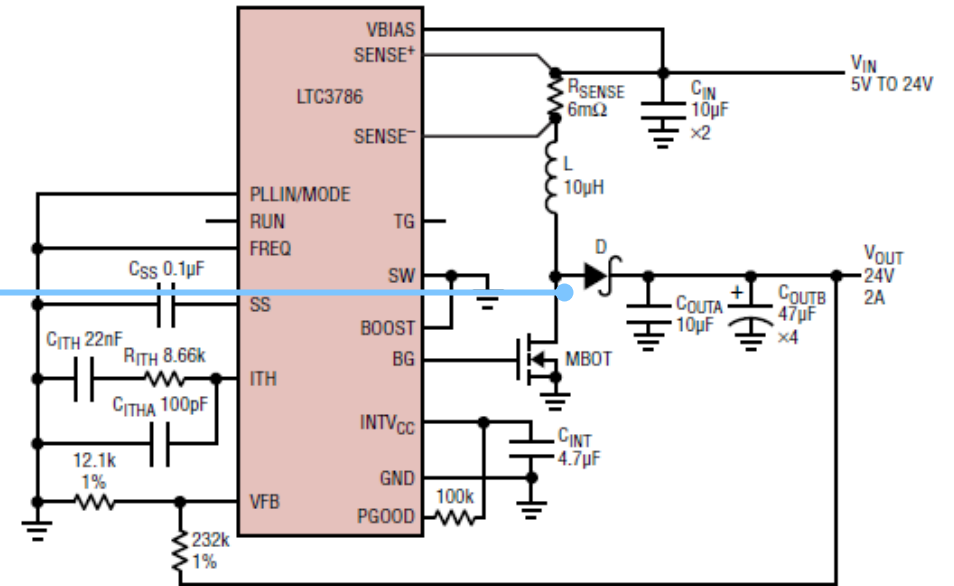
Max I_{OUT} with PWM
Mode selected



Boost Converters

Boost converters (Controllers)

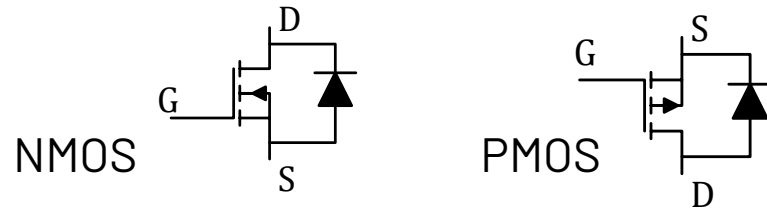
- Used for much higher power applications
- Can be non-synchronous (top switch is a Schottky diode) or Synchronous (FET)
- Diode drop doesn't greatly affect efficiency at high V_{OUT} . But if $I_{OUT} > 3A$, diode heating may become an issue
- Most Controllers have automatic Burst Mode operation to maximize light-load efficiency



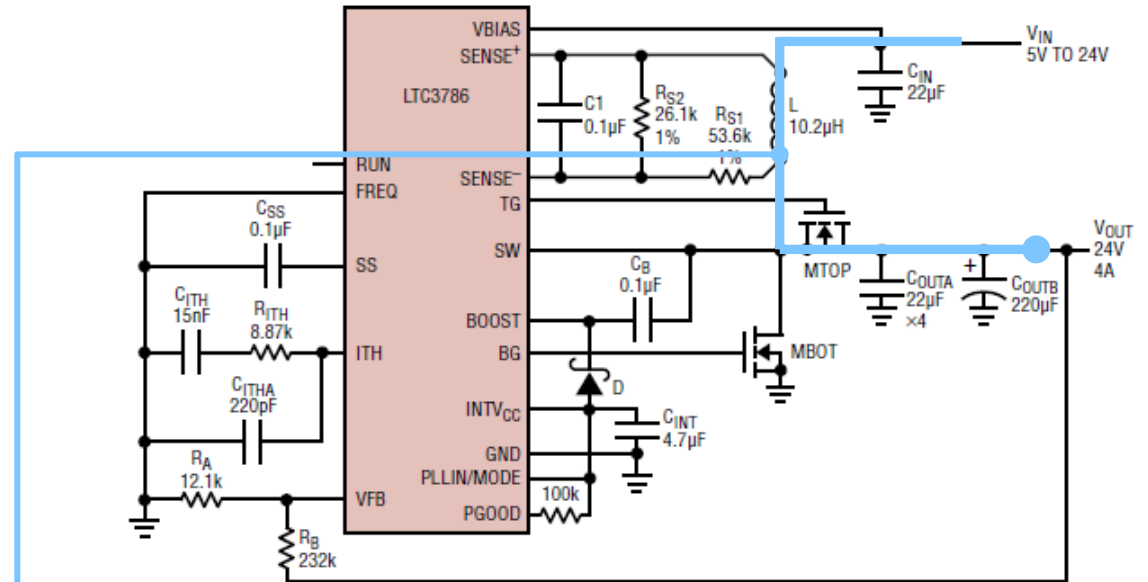
Boost Converters

Boost converters (Controllers)

- Note: All Silicon FETs have a body diode. This is a parasitic P-N junction between Source and Drain



- The Top FET (also called the Sync FET) will provide a path from V_{IN} to V_{OUT} through the body diode



Boost Converters

Boost converters – Compensation

- Many monolithic converters have internal compensation. Higher power monolithics may have external compensation.
- All Boost converters have a Right Half-plane Zero in the loop response, located at

$$f_{RHPZ} = \frac{R_{LOAD}}{2\pi L} * \left(\frac{V_{IN}}{V_{OUT}}\right)^2$$

Note that $R_{LOAD} = \frac{V_{OUT}}{I_{OUT}}$ and thus f_{RHPZ} depends on the load current, and f_{RHPZ} is at its lowest frequency at max load

Boost Converters

Boost converters – *RHPZ*

The *RHPZ* arises from the following facts:

1. When the output current rises, V_{OUT} droops a bit
2. The boost control circuit must increase the On-time to ramp the inductor current up to a higher value. But this leaves less Off-time for the current to be delivered to the load.
3. If the loop bandwidth is too high, the controller will keep increasing the On-time until it reaches its maximum, and the output voltage will collapse. This is an unstable condition.

The loop bandwidth should be $< 15\%$ of f_{RHPZ} to ensure stability

Boost Converters

Boost converters - RHPZ

Example: $V_{IN} = 12V$, $V_{OUT} = 24V$ @5A, $L=2.2\mu H$

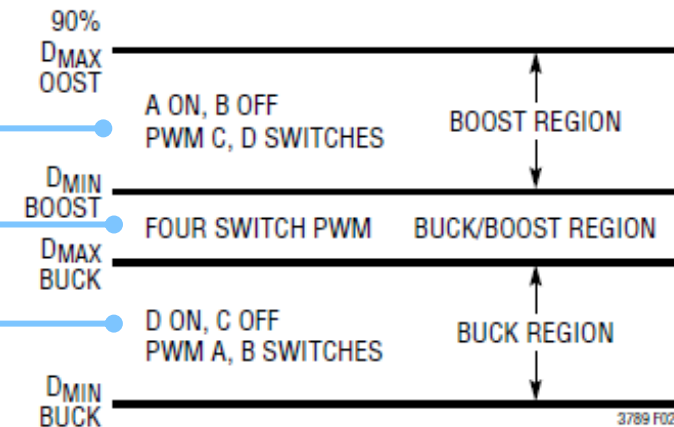
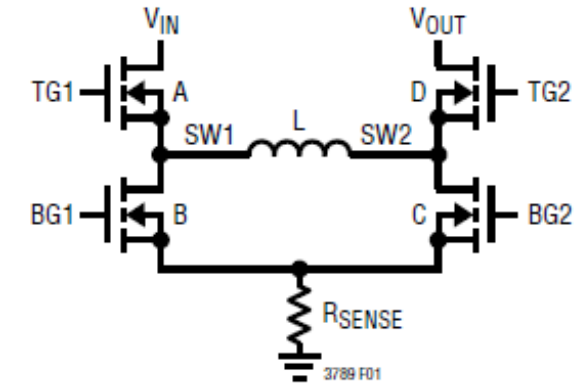
Calculate the RHPZ :

- $L = 2.2\mu H$, $R_{LOAD} = \frac{24V}{5A} = 4.8\Omega$
- $f_{RHPZ} = \frac{R_{LOAD}}{2\pi L} * \left(\frac{V_{IN}}{V_{OUT}}\right)^2 = \frac{4.8\Omega}{6.28 * 2.2\mu H} * \left(\frac{12V}{24V}\right)^2 = 86.9kHz$
- So, the Loop bandwidth should be $< 13kHz$
- LTPowerCAD example with LTC7804

Agenda

- 1 Overview
- 2 Boost Regulators
- 3 Buck-Boost Regulators**

- Can seamlessly transition from $V_{IN} < V_{OUT}$ to $V_{IN} = V_{OUT}$ to $V_{IN} > V_{OUT}$
- 4 Switches operate in Buck, Buck-Boost and Boost modes. Switches are labelled A, B, C, D
- Boost Mode ($V_{IN} < V_{OUT}$)
- Buck-Boost Mode (V_{IN} close to V_{OUT})
- Buck Mode ($V_{IN} > V_{OUT}$)



Buck-Boost Regulators

Buck-Boost Switchers

- Buck Mode ($V_{IN} > V_{OUT}$)
- Since R_{SENSE} is at the Source of Switch B, the controller operates in Valley Current Mode
- Switch C is OFF, Switch D is always ON
- Switches A, B toggle

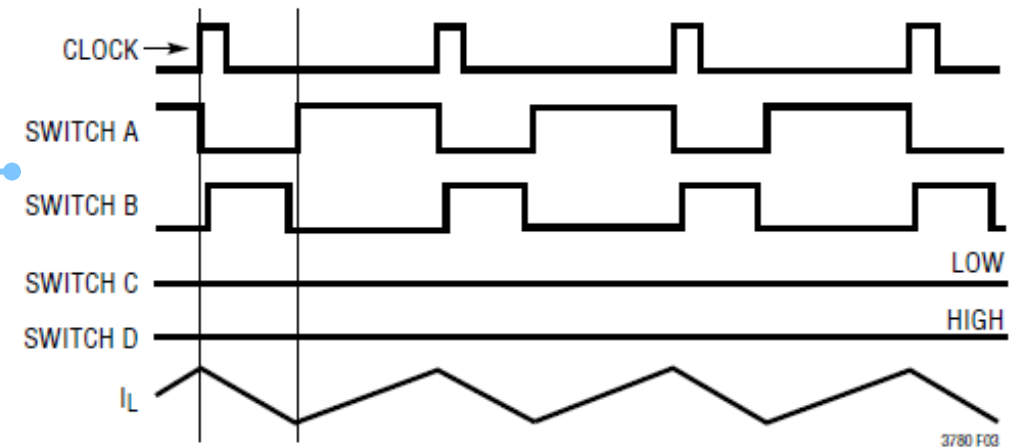
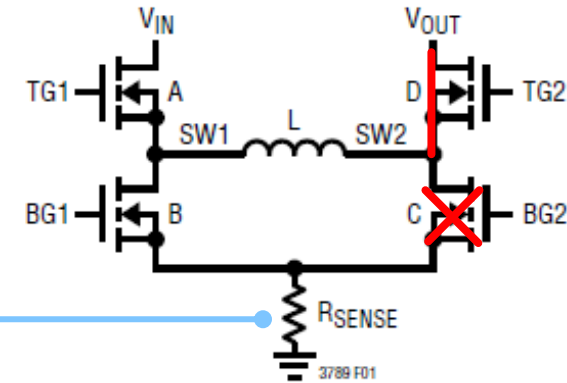


Figure 3. Buck Region ($V_{IN} \gg V_{OUT}$)

Buck-Boost Regulators

Buck-Boost Switchers

- Boost Mode ($V_{IN} < V_{OUT}$)
- Since R_{SENSE} is at the Source of Switch C, the controller operates in Peak Current Mode
- Switch B is OFF, Switch A is always ON
- Switches C, D toggle

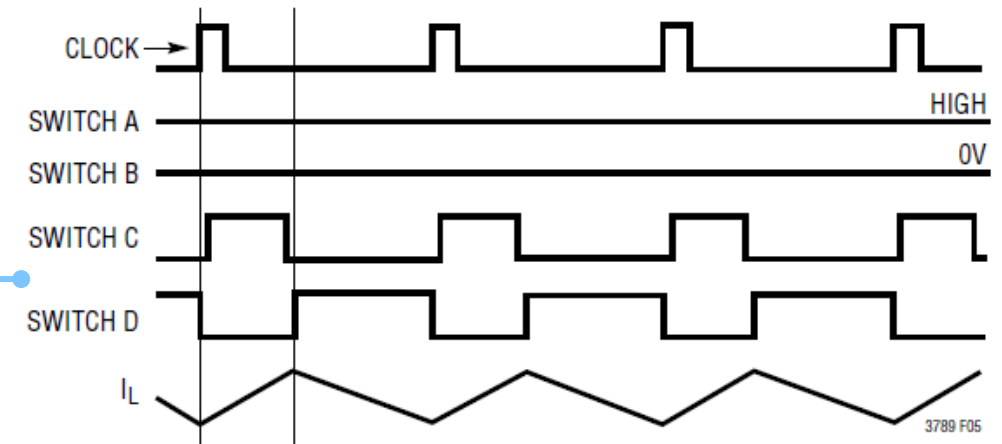
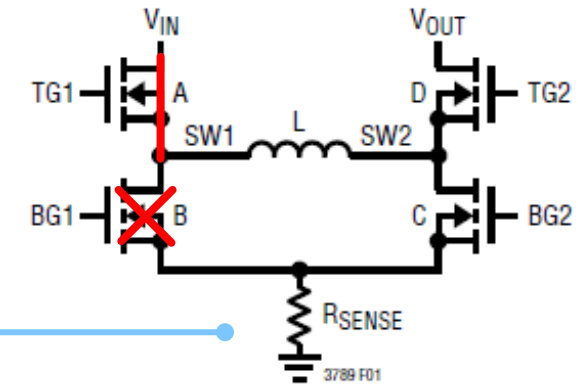
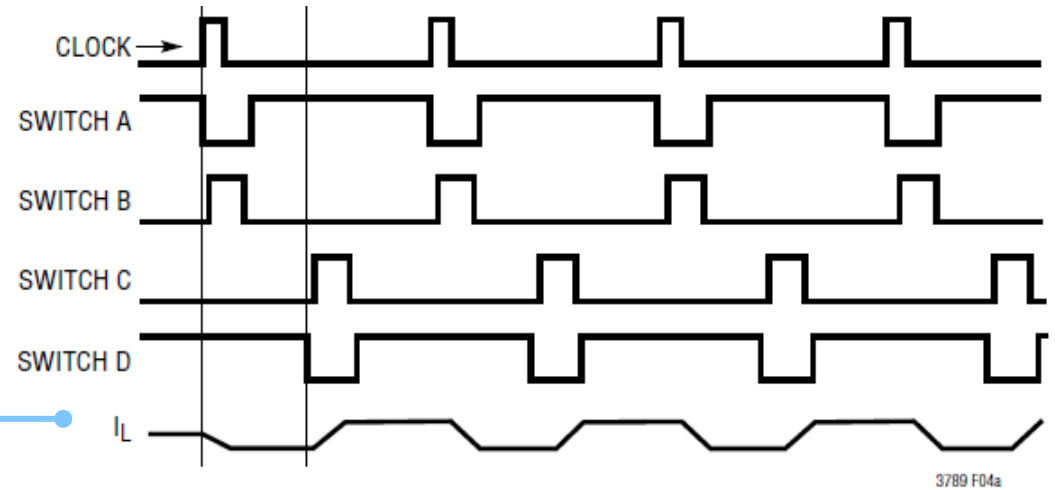
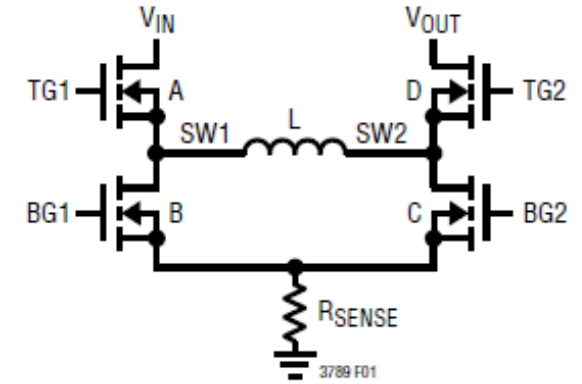


Figure 5. Boost Region ($V_{IN} < V_{OUT}$)

Buck-Boost Regulators

Buck-Boost Switchers

- Buck-Boost Mode (V_{IN} near V_{OUT})
- All 4 switches toggle. Typical operation is to have one Buck cycle followed by one Boost cycle

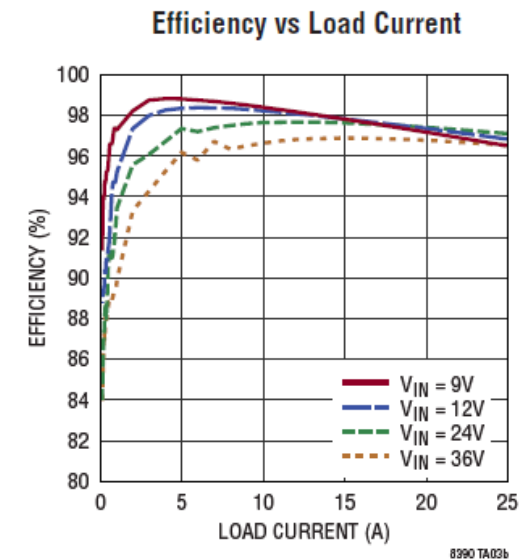
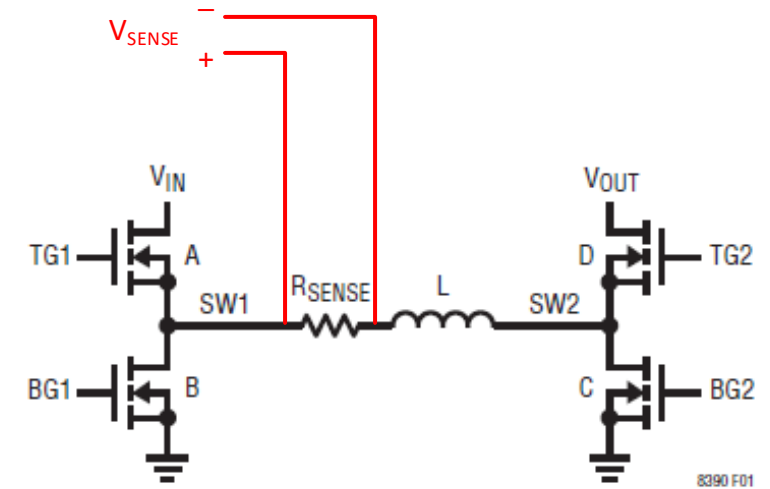


(4a) Buck-Boost Region ($V_{IN} \geq V_{OUT}$)

Buck-Boost Regulators

Buck-Boost Switchers

- Special Controller: LT8390
- Proprietary circuit allows current sensing with resistor in series with inductor.
- This means it can use Peak Current Mode control in all switching modes. This topology means that the Current Sense amplifier must be rail-to-rail and **FAST**
- The LT8390 also includes a second, rail-to-rail current sense amplifier for input or output current limiting (separate from inductor current sense).
- Efficiency: 12 V_{OUT} @25A



Boost Converters

Buck-Boost Switchers

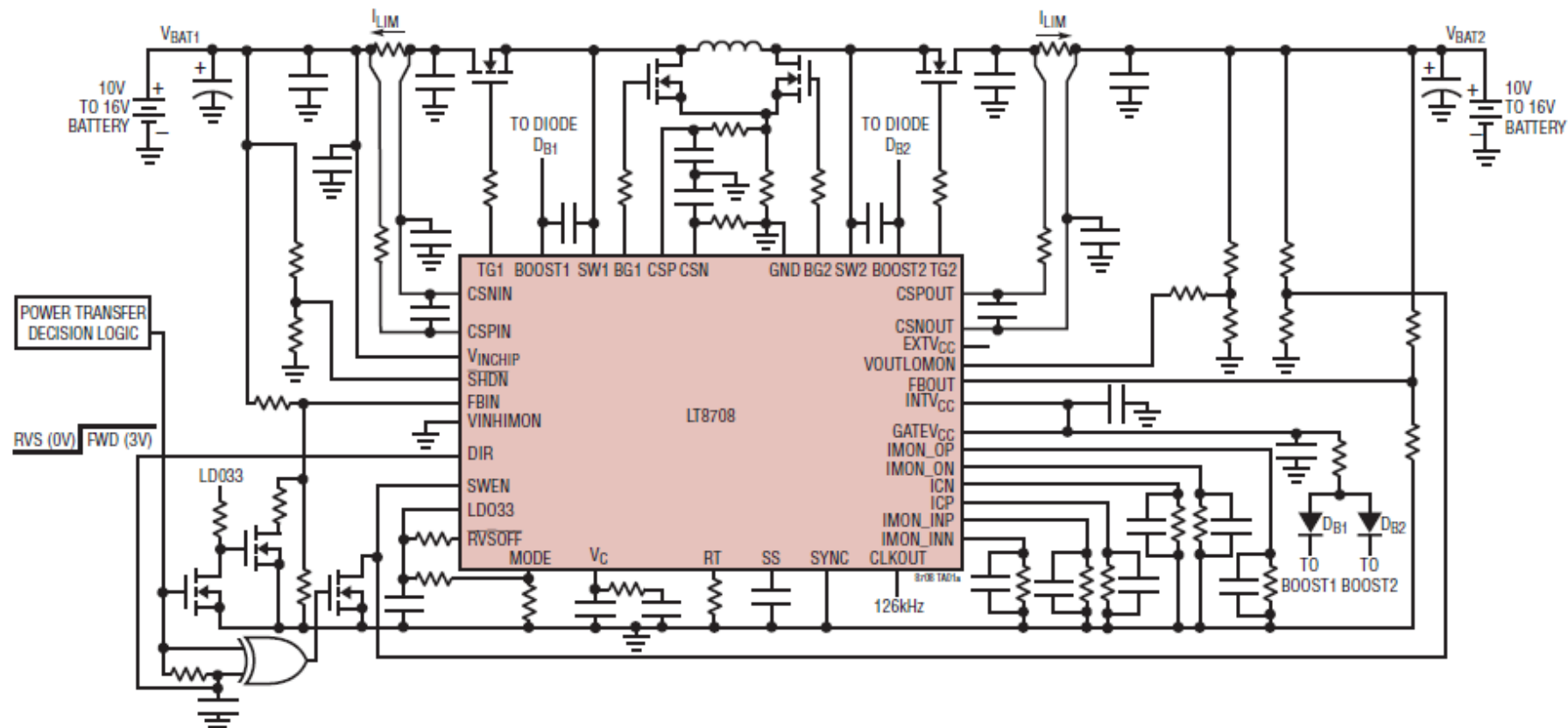
Bi-Directional switchers

- The LT8708 is a Buck-Boost controller which can run in Forward (conventional direction) or Reverse (backward direction) modes.
- This part is useful for dual-battery systems or other applications where power may need to be moved from Input to Out and vice-versa. Backup operation is another example of an application for a Bi-Directional regulator.
- This part has 6 control loops covering Input Current (forward & reverse), Output Current (forward & reverse), and V_{IN} and V_{OUT} .

Buck-Boost Regulators

Buck-Boost Switchers - Bidirectional

12V Bidirectional Dual Battery System with FHCM and RHCM



Boost Converters

Buck-Boost Switchers

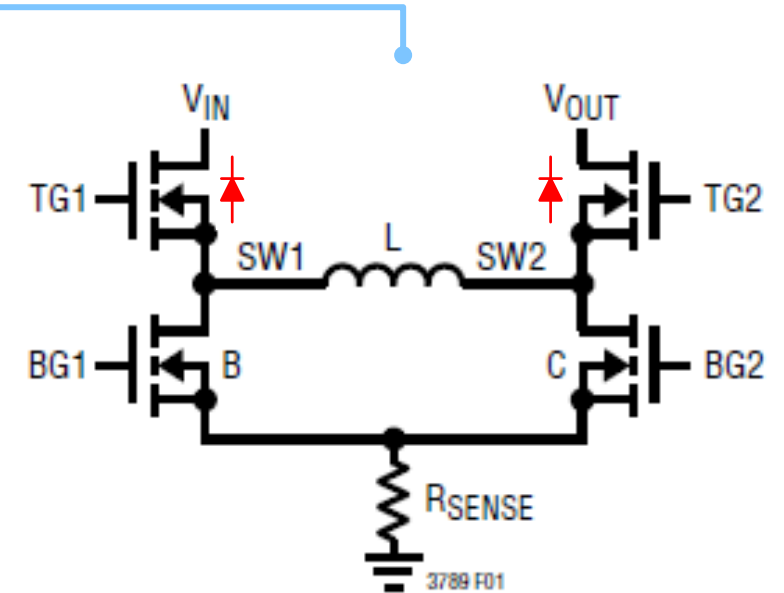
- Integration Levels
- Controllers for high power and/or high Voltage applications
- Monolithic regulators for switch currents up to ~7A and input Voltage up to ~40V
- Some monolithic parts have extremely low quiescent current (1 μ A or lower). These can be great for battery-powered or energy-harvesting applications.
- μ Modules for applications with limited board space and/or clients with limited Engineering resources or short timelines

Buck-Boost Regulators

Buck-Boost Switchers

Key Take-aways:

- Buck-Boost regulators can have very high efficiency. Above 97% is common when V_{OUT} is relatively large and V_{IN} is close to V_{OUT}
- Buck-Boost regulators have built-in Output Disconnect due to the body diodes in the switches. Useful when you need a pure Boost with Output Disconnect.
- The controller logic is a bit complicated, particularly when transitioning between modes. Some early adoptions of 4-switch Buck-Boost topologies have glitches or odd behaviour around the transition zones. ADI's Buck-Boost regulators will transition very smoothly across all the modes thanks to their advanced control loops.



The background is a solid blue color on the left, transitioning into a pattern of dark blue and medium blue triangles and cubes on the right. A large, light blue triangle points towards the right, partially overlapping the geometric pattern.

Thank you!

Q & A

AHEAD OF WHAT'S POSSIBLE

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